

Generation Interconnection System Impact Study Report

For Queue Project Q21-1740

East Berkshire 345 kV

October 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1740. The Interconnection Customer proposes a 786.6 MW natural gas combined cycle facility interconnecting at the East Berkshire 345 kV substation in the Tennessee area. The request was received on May 09, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 514 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,711,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wolf Hills Nergy 345 kV/138 kV XFMR
Contingency: Dashville / Spring Crk 345kV
Loading: 140.5% (Rating: 389 MVA)
DFAX: 45.8%
Pre-project loading: 48.0%
2. Facility: Wild Rose / Herrin 138kV
Contingency: Atoka / John Sevier 345kV L
Loading: 133.0% (Rating: 222 MVA)
DFAX: 26.5%
Pre-project loading: 39.0%
3. Facility: Palmer / N Bristol 138kV
Contingency: Dashville / Spring Crk 345kV
Loading: 124.4% (Rating: 427 MVA)
DFAX: 45.8%
Pre-project loading: 40.1%
4. Facility: Palmer / Wolf Hills Energy 138kV Line
Contingency: Spring Crk / Dashville 345kV
Loading: 122.9% (Rating: 445 MVA)
DFAX: 45.8%
Pre-project loading: 42.0%
5. Facility: King Substation / Herrin 138kV
Contingency: John Sevier / Atoka 345kV
Loading: 118.4% (Rating: 237 MVA)
DFAX: 26.5%
Pre-project loading: 30.2%
6. Facility: E Berkshire / Station A 138kV
Contingency: Palmer / Wolf Hills Energy 138kV
Loading: 114.7% (Rating: 233 MVA)

DFAX: 17.3%
Pre-project loading: 56.3%

7. Facility: Statio A / Drop 2 138kV line
Contingency: Wolf Hills Enrgy / Palmer 138kV
Loading: 112.1% (Rating: 228 MVA)
DFAX: 17.3%
Pre-project loading: 52.4%
8. Facility: George Dickie / Southcenter 138kV Line
Contingency: Kentucky Dam / Stock Islnd 138kV
Loading: 110.0% (Rating: 305 MVA)
DFAX: 16.4%
Pre-project loading: 67.7%
9. Facility: Drop 2 / Prine 138kV
Contingency: Palmer / Wolf Hills Energy 138kV
Loading: 108.5% (Rating: 224 MVA)
DFAX: 17.3%
Pre-project loading: 47.8%
10. Facility: Michigamme Falls Hy / Adairville 138kV
Contingency: E Berkshire 345/138 kV Autotransformer
Loading: 108.0% (Rating: 229 MVA)
DFAX: 10.1%
Pre-project loading: 73.3%
11. Facility: Cinder / Sprigg 138kV
Contingency: Neptune / Ragland 345kV line
Loading: 106.7% (Rating: 279 MVA)
DFAX: 12.4%
Pre-project loading: 71.7%
12. Facility: Sprigg / Kimper 138kV L
Contingency: Neptune / Ragland 345kV line
Loading: 106.3% (Rating: 288 MVA)
DFAX: 12.4%
Pre-project loading: 72.3%
13. Facility: Adairville / Warm Springs Tap 138kV
Contingency: E Berkshire 345 kV/138 kV Autotransformer
Loading: 103.1% (Rating: 225 MVA)
DFAX: 10.1%
Pre-project loading: 67.7%
14. Facility: Bluefield Avenue Substation / Inskip 138kV
Contingency: Lost Nation / No Troy 138kV
Loading: 100.7% (Rating: 223 MVA)
DFAX: 6.1%
Pre-project loading: 79.2%
15. Facility: Nassau / Inskip 138kV line
Contingency: No Troy / Lost Nation 138kV
Loading: 100.2% (Rating: 231 MVA)
DFAX: 6.1%
Pre-project loading: 79.5%
16. Facility: Kentucky Dam / Stock Island 138kV
Contingency: John Sevier / Atoka 345kV
Loading: 100.1% (Rating: 437 MVA)
DFAX: 12.4%

Pre-project loading: 77.9%

17. Facility: Abingdon / Warm Springs Tap 138kV Line

Contingency: E Berkshire 345-138 kV Transformer

Loading: 100.1% (Rating: 219 MVA)

DFAX: 10.1%

Pre-project loading: 63.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wolf Hills Energy 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$17,640,000	53% (\$9,331,000)
Wild Rose / Herrin 138kV	Rebuild line with higher-capacity conductor	\$2,381,000	51% (\$1,224,000)
Palmer / N Bristol 138kV	Rebuild line with higher-capacity conductor	\$2,640,000	53% (\$1,396,000)
Palmer / Wolf Hills Energy 138kV	Rebuild line with higher-capacity conductor	\$1,834,000	53% (\$970,000)
Herrin / King 138kV	Rebuild line with higher-capacity conductor	\$31,284,000	51% (\$16,088,000)
E Berkshire / Station A 138kV	Rebuild line with higher-capacity conductor	\$10,459,000	59% (\$6,170,000)
Drop 2 / Sta A 138kV	Rebuild line with higher-capacity conductor	\$1,203,000	59% (\$710,000)
Geoge Dickie / Southcenter 138kV	Rebuild line with higher-capacity conductor	\$3,460,000	22% (\$745,000)
Drop 2 / Prine 138kV	Reconductor line	\$3,823,000	59% (\$2,255,000)
Michigamme Falls Hy / Adairville 138kV	Reconductor line	\$598,000	64% (\$382,000)
Cinder / Sprigg 138kV	Reconductor line	\$1,174,000	24% (\$280,000)
Sprigg / Kimper 138kV L	Reconductor line	\$5,710,000	24% (\$1,362,000)
Warm Springs Tap / Adairville 138kV	Reconductor line	\$1,821,000	64% (\$1,163,000)
Inskip Substation / Bluefield Avenue 138kV	Reconductor line	\$5,020,000	15% (\$752,000)
Nassau / Inskip 138kV line	Reconductor line	\$2,515,000	15% (\$377,000)
Kentuckydam Substation / Stock Island 138kV line	Reconductor line	\$1,339,000	21% (\$285,000)
Abingdon / Warm Springs Tap 138kV Line	Reconductor line	\$1,912,000	64% (\$1,221,000)
	Total allocated to project		\$44,711,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.